Part 6 — Hosts: what each organism gives you, what it charges, and how to choose (weeks 58 to 61)

Part 5 ended with the observation that in four of five product classes the property deciding the outcome is not in the pathway: it is export, tolerance, accumulation physiology, membrane biology or gas transfer. All of those are host properties. So this part is written as a comparison against the five constraint sheets rather than as a set of species descriptions, and the question it keeps asking is the only one that matters at a host-selection meeting: which property decides this, and what do I give up by choosing on it?

Three pieces of arithmetic run through these chapters, and each is a calculation from earlier in the module appearing in a new currency. The compartment question in week 58 — whether putting a pathway in a peroxisome or mitochondrion pays — is part 3 chapter 48's dead-core problem with membranes instead of gel: a local concentration gain against a transport duty on every substrate and cofactor that crosses. The intracellular-product ceiling in week 59 is part 1 chapter 1's cell-volume argument, recomputed honestly at industrial cell density, where the answer turns out to be much less discouraging than the textbook version. And week 60's light penetration is the dead core a third time, with photons: light reaches a few millimetres into a dense culture and the rest of the vessel is dark, which is the photobioreactor's version of an oxygen gradient.

An instruction for how to use this part. The tracker asks you to build one host table and keep adding to it rather than duplicating it in each week. Chapter 61 contains that table, filled as far as this module can defend, with the cells marked where a number should be looked up rather than taken from me. Treat it as a template to fill with sources, not as a reference to cite.

Ch.	Week	Chapter
58	58	Yeast hosts: compartments, the lipid ceiling, and choosing between three yeasts
59	59	Gram-positive hosts: the envelope, the lysine arithmetic, and secretion as the deciding factor
60	60	Phototroph hosts: light as the limiting substrate, and the cost of taking the water out
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Yeast hosts: compartments, the lipid ceiling, and choosing between three yeasts

A eukaryote gives you folding, glycosylation, secretion, low-pH tolerance and organelles you can put a pathway into. It charges you a membrane between your enzymes and their substrates, and a toolkit that only exists in full for one of the three species.

Why a yeast at all

Four reasons, and it is worth separating them because different products need different ones. A eukaryote folds and glycosylates proteins a bacterium cannot — the reducing-cytoplasm problem of part 1 chapter 2 does not arise on a secretory route that passes through the endoplasmic reticulum. Yeasts tolerate low pH and high osmolarity, which matters for organic acids and for contamination control in a non-sterile process. Several have Generally Recognised As Safe status for food use, the host property part 5 chapter 54 noted cannot be engineered. And they have organelles, which is both the opportunity and the tax of this chapter.

The compartments to know: the nucleus; mitochondria, where pyruvate dehydrogenase sits and where the TCA cycle runs; the endoplasmic reticulum and Golgi, the secretory route where signal-peptide-bearing proteins fold, are glycosylated and are packaged into vesicles; peroxisomes, where β-oxidation of fatty acids happens in yeast exclusively — mammals run it in both mitochondria and peroxisomes, so a pathway ported from a mammalian source may expect the wrong compartment; the vacuole; and a cell wall of mannoproteins and glucan that is thick enough to matter for large secreted proteins.

Where acetyl-CoA lives, and why that is the first thing you engineer

This is the single most consequential fact about yeast as a chemical host. Pyruvate dehydrogenase is mitochondrial, so the acetyl-CoA it makes is mitochondrial. Fatty acid synthesis and the mevalonate pathway are cytosolic. **Acetyl-CoA does not cross the mitochondrial membrane**, so a cytosolic pathway cannot use the pool the cell makes most of, and the cytosolic pool has to be supplied by one of two routes:

- **The citrate shuttle with ATP-citrate lyase.** Citrate leaves the mitochondrion and is cleaved in the cytosol to acetyl-CoA and oxaloacetate, at a cost of one ATP per acetyl-CoA. This is the oleaginous yeasts' route and it is why they are oleaginous: *Y. lipolytica* has it, *S. cerevisiae* essentially does not.
- **The pyruvate dehydrogenase bypass.** Pyruvate is decarboxylated to acetaldehyde, oxidised to acetate, and activated to acetyl-CoA by acetyl-CoA synthetase — which cleaves ATP to AMP, so **two ATP equivalents** each, as part 1 chapter 4 insisted. This is *S. cerevisiae*'s route and it is expensive.

Hence the standard first intervention in any yeast strain for lipids, isoprenoids or polyketides: engineer cytosolic acetyl-CoA supply, by introducing ATP-citrate lyase, by replacing the bypass with a cheaper acetylating acetaldehyde dehydrogenase, or by rerouting through a phosphoketolase that avoids the decarboxylation entirely. Doing the terpene synthase first, before this, is the classic beginner's order and it is why so many first attempts give milligrams.

The lipid ceiling, computed

Palmitate, $C_{16}H_{32}O_2$, is eight acetyl-CoA condensed with seven rounds of malonyl-CoA chemistry. The bill:

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8 acetyl-CoA = 4 glucose          carbon: 16 of 24 → 67%, so a carbon-only ceiling of 0.356 g/g
7 ATP  for acetyl-CoA carboxylase (malonyl-CoA)
8 ATP  for ATP-citrate lyase, one per acetyl unit shuttled    → 15 ATP per palmitate
14 NADPH for the seven reduction cycles (two each)
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Now pay for the NADPH, which is the step most back-of-envelope calculations skip. Routed through a fully cycling oxidative pentose phosphate pathway, one glucose yields 12 NADPH while releasing all six carbons as CO_2, so 14 NADPH costs 1.17 glucose. Total feedstock is 4 + 1.17 = **5.17 glucose per palmitate**, and the ceiling falls from 0.356 to **0.275 g/g**. For triacylglycerol, with its glycerol backbone, the same assumptions give about 0.28 g/g. Published ceilings for TAG on glucose sit a little higher, around 0.30 to 0.33 g/g, because some NADPH comes from malic enzyme or from a partially cycling pentose phosphate pathway that is less wasteful of carbon than my assumption — the difference between my number and theirs is entirely in how the reductant is sourced, which is exactly the point.

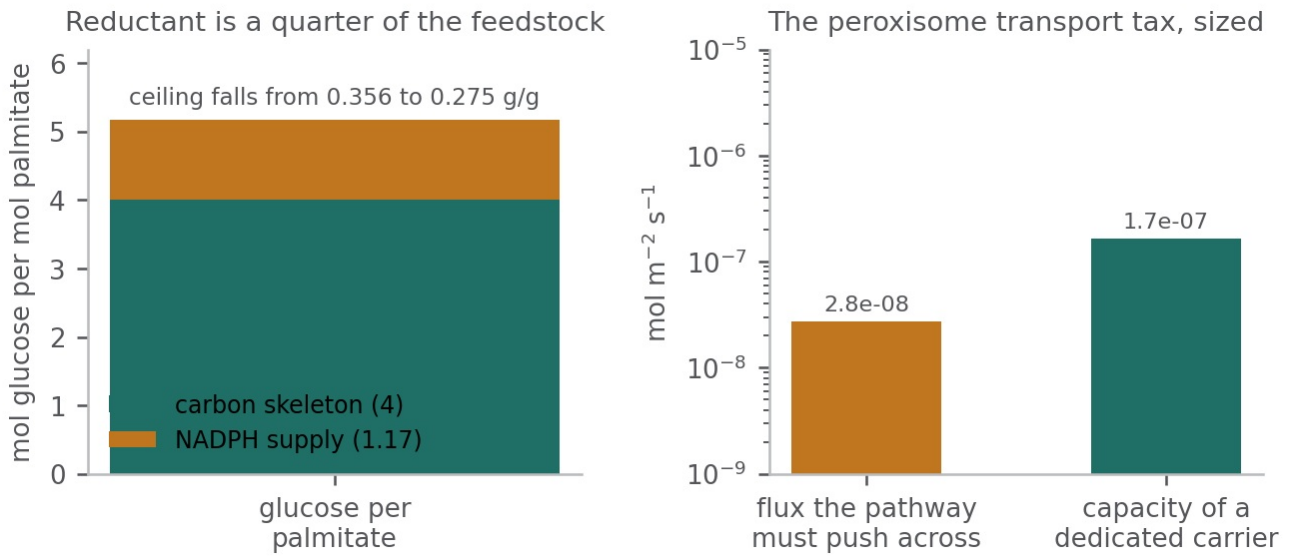


Figure 58.1 Left: the glucose bill for one palmitate, split into carbon skeleton and the glucose burned to supply NADPH — reductant is nearly a quarter of the feedstock, and it drops the ceiling from 0.356 to 0.275 g/g. Right: the peroxisome transport duty. A cell making 0.5 g/L/h at 20 gDCW/L must push about 2.8×10^{-8} mol·m^{-2}·s^{-1} across the organelle membrane, against roughly 1.7×10^{-7} for a carrier at 1,000 copies per µm² and k_{cat} 100 s^{-1} — feasible, with about sixfold margin, and only with a dedicated transporter. Computed for this module from the stated assumptions.

Three consequences follow, and they answer the week's questions directly. **Overexpressing fatty acid synthase alone rarely raises titer** for two reasons at once: part 2 chapter 15's control analysis says a single enzyme holds a fraction of the control, and the balance above says the binding constraints are cytosolic acetyl-CoA and NADPH, neither of which the synthase supplies. **The carbon-to-nitrogen ratio is the switch**: with nitrogen exhausted and carbon still fed, protein synthesis stops, AMP deaminase activity reduces the AMP pool, mitochondrial isocitrate dehydrogenase slows, citrate accumulates and is exported to the cytosol — which is the lipid-accumulation trigger, and it is set in a fed-batch by the ratio of carbon to nitrogen in the feed, not by a gene. And **the ATP bill is real**: 15 ATP per palmitate is paid by respiring carbon that cannot become product, which is part of why observed yields sit well below even 0.275.

Compartment engineering, with the transport tax sized

Putting a pathway into an organelle does two good things: it concentrates the enzymes and their intermediates in a small volume, and it keeps a toxic intermediate away from the cytosol. A peroxisome of 0.2 µm radius has a volume of about 0.034 fL against a yeast cell's 50 fL, so a cell with ten of them confines the pathway to under 1% of its volume — a hundredfold concentration gain for the same number of enzyme molecules, which for an enzyme operating below its K_M is a hundredfold rate gain, by part 1 chapter 3's arithmetic.

The tax is that every substrate, every cofactor and the product must cross a membrane. Size it as the figure does: a cell making 0.5 g/L/h of a 150 Da product at 20 gDCW/L carries about 1.4×10^{-20} mol·s^{-1} per cell, which across one peroxisome's 5×10^{-13} m² of membrane is a flux density of 2.8×10^{-8} mol·m^{-2}·s^{-1}. A dedicated carrier at 1,000 copies per µm² with a turnover of 100 s^{-1} delivers about 1.7×10^{-7} — six times the requirement, so it works, but only with a transporter expressed at high density and only for the species that has one. There is no carrier for your novel intermediate, and NADPH does not cross at all: the organelle must regenerate its own, which means installing a dehydrogenase and a source of its substrate inside the compartment.

So the honest rule, which is the same conclusion as the dead-core calculation: compartmentalisation pays when the intermediate is toxic or the precursor is already made inside that compartment — the isobutanol-in-mitochondria case of part 5 chapter 55 — and loses when it merely sounds elegant, because you have added three transport problems to solve a concentration problem that a stronger promoter might have solved.

Three yeasts, compared properly

	S. cerevisiae	*Y. lipolytica*	*K. phaffii*
Substrate range	Glucose, sucrose, maltose; not methanol; poor on alkanes	Glucose, glycerol, acetate, alkanes and fatty acids — hydrophobic substrates are a speciality	Glycerol and methanol; glucose less favoured industrially because the strong promoters are methanol-linked
Crabtree status	Positive: makes ethanol from sugar even with oxygen, so carbon leaks to ethanol at high flux	Negative: respires sugar fully, which is why high-density aerobic growth is straightforward	Negative

Lipid storage	Little — a few per cent of dry weight; engineering it for lipid means rewiring acetyl-CoA supply and storage	Oleaginous: tens of per cent of dry weight natively, with the citrate shuttle already present	Low
Secretion	Moderate; hyperglycosylates	Strongly secretory, with native lipases and proteases exported at high titer	Excellent — grams per litre with a clean extracellular background, which is why it dominates protein production
Tool maturity	The best toolkit in biology: markers, promoter libraries, efficient homologous recombination, CRISPR, synthetic chromosomes	Good and improving; non-homologous end joining dominates, so targeting is harder	Good for expression, thinner for pathway work
GRAS / food use	Yes, long history	Yes for several products and processes	Yes, with a history in food enzymes — the soy leghemoglobin precedent
Documented products	Ethanol, artemisinic acid, farnesene, resveratrol, insulin precursors	Lipids and oleochemicals, citric acid, erythritol, omega-3 oils, cosmetic ingredients	Secreted enzymes and proteins; soy leghemoglobin

One line each. *S. cerevisiae* is uniquely good at being engineered and uniquely bad at keeping carbon out of ethanol. *Y. lipolytica* is uniquely good at lipid and at hydrophobic substrates and uniquely awkward to target genetically and to keep morphologically consistent, since it is dimorphic and switches between yeast and hyphal forms with conditions — which changes viscosity, oxygen transfer and your optical density calibration at the same time. *K. phaffii* is uniquely good at secreting protein at high cell density and is not where you would build a twelve-gene small-molecule pathway.

Tools, and the three things that do not carry across

DNA delivery and maintenance differ in ways that matter for a project plan. In *S. cerevisiae*, episomal plasmids with centromeres or 2-micron origins are routine and homologous recombination is efficient enough that a 40 bp flank targets a locus — which is why assembly in yeast is itself a cloning technique. In *Y. lipolytica*, non-homologous end joining outcompetes homologous recombination, so targeting a locus needs long flanks, a *ku70*-deleted background, or a CRISPR cut to force repair at the right place. In *K. phaffii*, integration is the norm and expression is usually driven from the methanol-inducible promoter system, with methanol-free variants now available.

Three things that work in *S. cerevisiae* and do not simply carry across: the promoter library, because promoter strength does not rank the same way in another yeast and must be re-measured; the auxotrophic marker set, because the corresponding mutants may not exist in the new host and must be made; and recombination-based assembly in the host itself, which depends on the efficient homologous recombination the other two lack. The first three experiments to run in a yeast new to you follow from exactly that: measure growth rate and the cell-density-to-dry-weight calibration in your medium; test DNA delivery and quantify transformants per microgram with a known construct; and measure a small promoter and terminator set against a fluorescent reporter so that you have numbers rather than someone else's ranking.

Host sheet — the yeasts

Choose a yeast when	The product needs eukaryotic folding, glycosylation or a secretory route; when a P450 needs an endoplasmic reticulum; when low pH or high osmolarity is part of the process; when GRAS status is required; or when lipid or hydrophobic-substrate metabolism is the point.
The first thing to engineer	Cytosolic acetyl-CoA supply — via ATP-citrate lyase, a cheaper bypass, or phosphoketolase — before any pathway enzyme.
Ceiling to quote	Lipid on glucose: 0.356 g/g on carbon alone, 0.275 g/g once NADPH is paid for from the pentose phosphate pathway, with 15 ATP per palmitate on top.
The switch	Carbon-to-nitrogen ratio, set in the feed, not a gene.
Compartments	Pay when the intermediate is toxic or the precursor is already inside; the tax is a transporter for every species crossing, and NADPH must be regenerated locally.
Honest bottlenecks	Glycosylation that differs from the human pattern; proteases in the medium; plasmid instability; Crabtree overflow in *S. cerevisiae*; dimorphism in *Y. lipolytica*; and a toolkit that does not transfer between the three.

Three things people get wrong

- ‘Yeast has acetyl-CoA, so a cytosolic pathway can use it.’ Most of it is mitochondrial and does not cross; the cytosolic pool costs one or two ATP per unit to make.
- ‘Compartmentalisation concentrates the pathway, so it helps.’ It also adds a transport duty on every substrate and cofactor, and NADPH does not cross at all.

- ‘The *S. cerevisiae* toolkit carries over to other yeasts.’ Promoters, markers and recombination-based methods all have to be rebuilt.

Self-check — answer on paper first

1. Why does a cytosolic isoprenoid pathway in *S. cerevisiae* cost more ATP than the same pathway in *Y. lipolytica*?

Because the cytosolic acetyl-CoA comes from different routes. *S. cerevisiae* uses the pyruvate dehydrogenase bypass, whose acetyl-CoA synthetase step cleaves ATP to AMP and therefore costs two ATP equivalents per acetyl unit; *Y. lipolytica* has an active citrate shuttle with ATP-citrate lyase, which costs one. For a C15 terpene that is nine acetyl units, so eighteen ATP equivalents against nine — a difference that has to be met by respiring carbon, and therefore appears in the yield.

2. Compute the glucose required per mole of palmitate if half the NADPH comes from a source that costs no carbon, and say what that does to the ceiling.

Seven of the fourteen NADPH then cost 7/12 = 0.58 glucose, so the total is 4.58 glucose and the ceiling rises from 0.275 to 256.42/(4.58 × 180.16) = 0.311 g/g. That is roughly the published range for TAG, and it shows that the entire difference between my number and the literature's is the assumption about reductant sourcing — which is why the first question to ask of any lipid-yield claim is where the NADPH was assumed to come from.

3. A colleague proposes moving a six-enzyme pathway into the peroxisome to raise its local concentration. What do you ask?

What crosses the membrane, and how. Specifically: which substrate enters and is there a native transporter for it, or must one be found; whether the pathway consumes NADPH or ATP, since neither crosses and both would have to be regenerated inside; how the product leaves; and whether the peroxisomal targeting signal actually imports folded multi-subunit enzymes, which it does for some and not others. Then the comparison question: is the intermediate toxic, or is the precursor already made in there? If neither, the concentration gain is probably obtainable more cheaply with expression tuning in the cytosol.

What to open a book for. No single book. The literature searches the tracker names are the sources — *Y. lipolytica* as a lipid host, cytosolic acetyl-CoA supply, *K. phaffii* secretion, and subcellular compartmentalisation in yeast. Part 1 chapters 1 and 4 of this module cover the compartment and ATP-equivalent arithmetic, and part 3 chapter 48 is the transport analogy worth rereading before the compartment discussion.

Gram-positive hosts: the envelope, the lysine arithmetic, and secretion as the deciding factor

One membrane instead of two. That single structural fact decides what these hosts are for: a secreted protein crosses one barrier and arrives in the medium, and there is no endotoxin to remove afterward.

The envelope, and what it changes

Draw both. *E. coli*: inner membrane, then a periplasm containing the peptidoglycan layer, then an outer membrane whose outer leaflet is lipopolysaccharide. A Gram-positive: one cytoplasmic membrane, then a thick peptidoglycan wall, then the medium. A signal peptide in *E. coli* delivers a protein to the periplasm, and getting it to the medium needs a second, separate export step; in a Gram-positive the same signal peptide delivers it to the outside of the single membrane, from which it diffuses through the wall into the medium.

Three consequences. **Secretion is simpler and higher-yielding**, which is why *B. subtilis* and its relatives make most of the world's industrial enzymes. **There is no lipopolysaccharide**, so no endotoxin to remove — a significant advantage for any injectable product, and the reason endotoxin removal is a line item in *E. coli*-derived biologics and not in Gram-positive ones. And **the periplasm is gone**, which is occasionally a loss: it is a small oxidising compartment where disulfide bonds form and where a product can be concentrated and protected, and some *E. coli* processes depend on it.

What the wall still makes difficult: it is a thick crosslinked mesh, and very large or slowly-folding proteins can be retained or degraded in transit through it. Lysis for an intracellular product is also harder than in *E. coli*, since the wall resists mechanical and enzymatic disruption — a cost that only appears if you choose a Gram-positive and then keep the product inside.

Secretion as the deciding factor, with the cell-volume arithmetic redone

Part 1 chapter 1 computed that an intracellular product is capped by cell volume, using 10^9 cells per millilitre and arriving at 0.1% of the broth volume and therefore a titer of a fraction of a gram per litre. That calculation is correct and it is also, for an industrial process, pessimistic — and the week asks you to redo it at a realistic density. Take about 2 mL of cell volume per gram of dry weight:

Cell density	Cell volume fraction	Broth titer at 200 g/L inside the cells
1 gDCW/L (the textbook case)	0.2%	0.4 g/L
10 gDCW/L	2%	4 g/L
50 gDCW/L	10%	20 g/L
100 gDCW/L (high-cell-density fed-batch)	20%	40 g/L

So an intracellular product at very high cell density is not hopeless — 20 to 40 g/L is a real process, which is exactly why intracellular products like PHA work at all, as part 5 chapter 56 showed. The honest statement is the combined one: **the cell-volume ceiling scales with cell density, so an intracellular product forces you into high-cell-density operation**, with everything that implies for oxygen transfer, cooling and feed control, and then forces lysis, debris removal and a harder purification. A secreted product needs none of that.

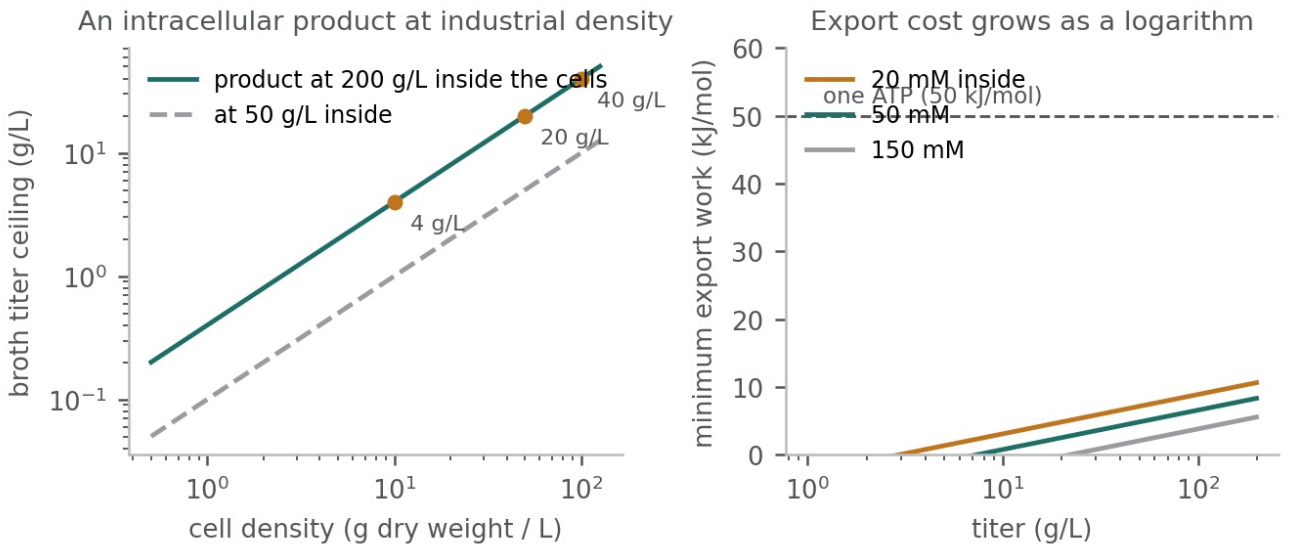


Figure 59.1 Left: the intracellular titer ceiling against cell density, at two intracellular concentrations — the textbook 0.1% case sits at the far left. Right: the thermodynamic minimum work of exporting a product against its own gradient, which grows only as the logarithm of titer: from 50 to 150 g/L of lysine it rises from 4.8 to 7.6 kJ/mol, a tenth to a seventh of an ATP. Computed for this module; 2 mL of cell volume per gram dry weight and a 50 mM intracellular concentration are the assumptions.

The lysine case, with the numbers that decide it

C. glutamicum makes glutamate and lysine at the million-tonne scale, and the lysine arithmetic is the best single illustration in this module of parts 1, 2 and 5 working together.

The ceiling. Reported theoretical maxima are 0.81 mol per mol of glucose for the native network and 0.86 with a transhydrogenase — which in mass terms is **0.657 and 0.698 g/g**. Note what moves it: not carbon, but a cofactor device. Lysine needs four NADPH per molecule, the oxidative pentose phosphate pathway supplies them at the cost of one carbon per two NADPH, and a transhydrogenase shifts part of that demand onto the NADH that glycolysis already makes — which is part 2 chapter 19's three-way choice, appearing as a 6% difference in the ceiling of a real industrial product.

The tension, stated properly. Carbon yield and reductant supply pull against each other: every glucose sent through the oxidative branch for NADPH loses a carbon as CO_2, and every glucose kept in glycolysis for carbon fails to supply NADPH. The literature addresses it three ways, and you should be able to name them: engineer a transhydrogenase, so NADH substitutes for NADPH at an energy cost; replace a glycolytic dehydrogenase with an NADP-dependent one so that glycolysis itself makes NADPH, which is the engineered NADPH-generating glycolysis the week's sources describe; or change the pathway enzymes' cofactor preference toward NADH. All three are cofactor-balance interventions rather than pathway interventions, which is the pattern.

Export, and whether it is a real term. The minimum work of moving lysine out against its own gradient is $RT \cdot \ln(c_{out}/c_{in})$: with 50 mM inside, 4.8 kJ/mol at 50 g/L, 6.6 at 100 g/L and 7.6 at 150 g/L — a tenth to a seventh of one ATP equivalent. So the thermodynamic term is small and, because it is logarithmic, **barely grows with titer**: tripling the titer adds 60% to a term that was already small. What is not negligible is the practical cost: real carriers are not reversible, so budget something closer to one ATP or one to two protons per molecule, which at 150 g/L is a few per cent of the cell's energy budget and belongs in the balance rather than in a footnote. The useful conclusion for a design discussion is that export energy is a real but second-order cost, and that the reason *C. glutamicum* wins is not the energy but that the carriers exist at all.

B. subtilis, and what its proteases do to your product

B. subtilis dominates industrial enzyme production because it secretes native amylases and proteases at commercial titers, has excellent classical and modern genetics, and is naturally competent — DNA uptake needs no electroporation, which makes library work unusually easy. It also secretes, natively, a set of extracellular proteases whose job is to degrade extracellular protein, and your heterologous product is extracellular protein.

That is why protease-deficient backgrounds exist, with several of the extracellular protease genes deleted. What they buy is a product that survives in the medium; what they cost is slower growth, some nutritional fastidiousness since the proteases also feed the cell, and sometimes a strain that sporulates or lyses differently. The failure mode to recognise is specific and the week asks for it directly: **a secreted protein that appears in the medium and then disappears is being degraded**, and the first checks are a protease-deficient background, a time course with protease inhibitors added

to samples, and a zymogram or activity assay on the spent medium to confirm proteolysis rather than aggregation or adsorption.

The other failure modes, with what to monitor: sporulation, which diverts part of the culture irreversibly late in a fed-batch — monitor by phase-contrast microscopy and by the sudden plateau in biomass; phage, which in a plant is a serious event rather than a nuisance; foaming from the secreted protein itself, which is worse the better your secretion works; and plasmid loss, which is why integration is standard for production strains.

Three experiments that tell you whether a new product can be secreted at all: express it with two or three different signal peptides, since signal-peptide performance is product-specific and is the single largest variable; run a time course measuring the product in the medium and in the cell fraction, to distinguish a secretion failure from a degradation one; and test it in a protease-deficient background, which separates the two remaining explanations at a stroke.

Host sheet — the Gram-positives

Choose one when	The product is a secreted protein, or an amino acid or other charged metabolite needing a native exporter; when endotoxin must be avoided; when GRAS status and a food or feed market matter.
Decisive property	One membrane: secretion crosses a single barrier and the product is harvested from the medium, with no lysis, no debris and no endotoxin removal.
Ceiling to quote	Lysine 0.81 mol/mol native, 0.86 with a transhydrogenase — 0.657 and 0.698 g/g. The difference is a cofactor device, not carbon.
Export cost	Thermodynamic minimum 5–8 kJ/mol across 50–150 g/L; practically about one ATP per molecule. Real, second-order, and logarithmic in titer.
Characteristic bottlenecks	Extracellular proteases degrading the product; sporulation; the wall limiting large proteins; phage; foaming from your own secreted protein.
Wrong choice when	The pathway needs eukaryotic folding or glycosylation, or the process depends on the oxidising periplasm that *E. coli* provides.

Three things people get wrong

- ‘An intracellular product is capped at a fraction of a gram per litre.’ At 1 gDCW/L, yes. At 100 gDCW/L the cells are a fifth of the broth and the cap is 40 g/L.
- ‘Export energy is negligible.’ The thermodynamic term is, and the carrier's practical cost of about one ATP per molecule is a few per cent of the budget at high titer.
- ‘*B. subtilis* secretes well, so my protein will accumulate.’ It also secretes proteases whose function is to degrade extracellular protein.

Self-check — answer on paper first

1. Explain why secretion changes the downstream process, in the order the unit operations occur.

For a secreted product: separate cells from broth by centrifugation or filtration, discard the cells, and purify from a relatively clean liquid. For an intracellular one: concentrate the cells, lyse them mechanically or enzymatically, remove the debris — which is fine, compressible and the single hardest solid–liquid separation in the sequence — and then purify from a solution containing the entire soluble contents of the cell, including nucleic acids that raise the viscosity and host proteins that must be resolved from the product. The secreted route removes two unit operations and starts the purification from a far simpler feed, which is usually worth more than any titer difference.

2. Why does a transhydrogenase raise the lysine ceiling by only about 6%, and why is that worth having?

Because it does not create reductant, it converts NADH to NADPH at an energy cost, so it substitutes a cheaper source for part of a demand that would otherwise be met by the oxidative pentose phosphate pathway at the cost of one carbon per two NADPH. Only the carbon saved shows in the yield, and only part of the demand can be shifted before energy becomes limiting — hence 0.81 to 0.86 mol/mol. It is worth having because in a million-tonne commodity made at a few hundred dollars a tonne, six per cent of the yield is an enormous sum, and because the intervention is a single gene.

3. A secreted enzyme reaches 1.2 g/L at 20 h and 0.3 g/L at 36 h. Give your diagnosis and the three experiments.

Proteolysis in the medium, most likely, since the loss is of product already successfully secreted. The experiments: repeat in a protease-deficient background, which is the definitive test; add protease inhibitors to samples at harvest and compare, which distinguishes degradation in the vessel from degradation during handling; and assay the spent medium for protease activity directly, ideally with your own product as substrate. If none of those explains it, the remaining candidates are aggregation or adsorption to cells and vessel surfaces, which a mass balance on the cell pellet and a

wash of the vessel would catch.

What to open a book for. Brock's cell-envelope chapter, which is genuinely the right source for the structural comparison and is worth reading with both envelopes drawn side by side. Then the reported theoretical maxima for lysine in *C. glutamicum* and the engineered NADPH-generating glycolysis paper in Appl. Environ. Microbiol. for the cofactor arithmetic, plus the literature on amino-acid exporters and on protease-deficient *B. subtilis* strains. Part 2 chapter 19 is the chapter of this module to have open beside them.

Phototroph hosts: light as the limiting substrate, and the cost of taking the water out

Light is a substrate you cannot stir. It arrives at a surface, is absorbed within millimetres, and leaves most of your vessel dark — which is why the unit is grams per square metre per day and why the water is the problem.

Light as the substrate, and the depth it reaches

Attenuation through a culture follows Beer–Lambert, with the biomass itself as the absorber:

$$I(z) = I_0 \cdot \exp(-k \cdot X \cdot z) \qquad k \approx 0.15\ \text{m}^2 \text{ per gram of biomass, } X \text{ in g/L, } z \text{ in metres}$$

Put numbers in and the problem is immediate. At a specific attenuation coefficient of 0.15 m²/g — a middle value for microalgae, and the assumption that matters most here — light falls to 10% of its surface value at the following depths:

Biomass density	Depth at 10% of surface light	Depth at 1%	Lit fraction of a 30 cm pond
0.25 g/L	61 mm	123 mm	20%
0.5 g/L	31 mm	61 mm	10%
1 g/L	15 mm	31 mm	5%
5 g/L	3 mm	6 mm	1%

At a density that would be considered dilute in any heterotrophic fermentation — one gram per litre, against a hundred in a fed-batch — light penetrates fifteen millimetres and 95% of a pond is dark. This is the structural fact of the field, and the comparison to make is with part 3 chapter 48's dead core: in both cases a substrate is consumed faster than it penetrates, leaving most of the catalyst idle. The difference is that oxygen can be raised by pressure and glucose can be fed, while photons arrive at a fixed areal rate set by the sun, and mixing cannot deliver more of them — it can only move cells between the lit layer and the dark one.

Which is why the unit is areal. You are not buying litres; you are buying illuminated area, and therefore land or glazing. **Grams per square metre per day is the only honest productivity unit for a phototroph**, and a paper reporting grams per litre per day for a thin photobioreactor is reporting a number that does not scale, because the thinness is what produced it.

What a pond can actually produce

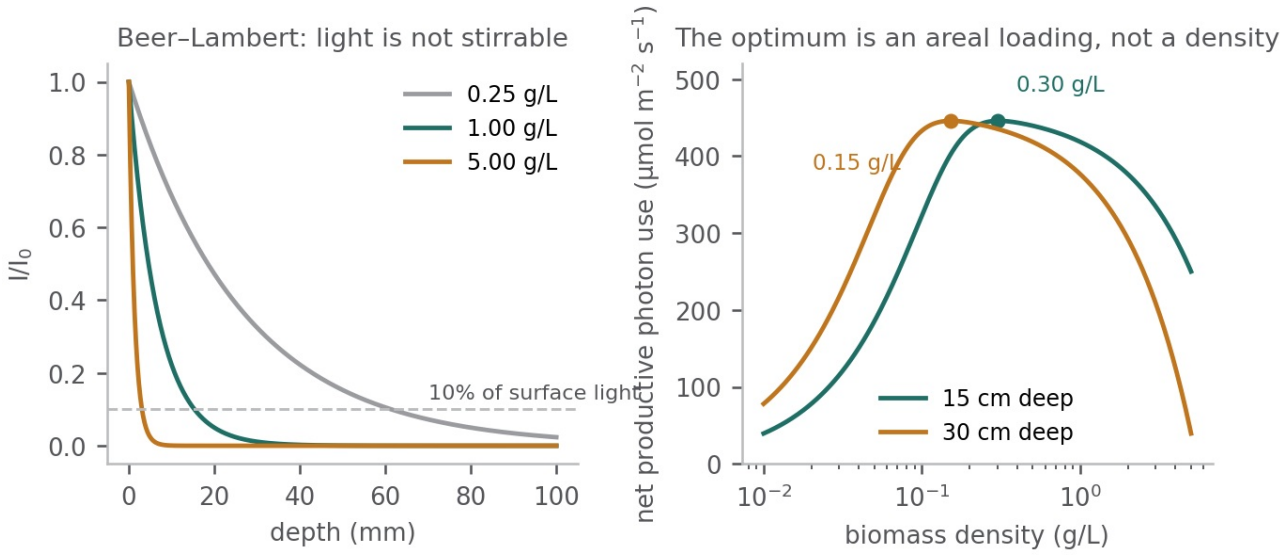


Figure 60.1 Left: Beer–Lambert attenuation at three densities, showing where the 10% depth falls. Right: net productive photon use against biomass density for two pond depths, with saturation wasting photons at high local irradiance and maintenance respiration consuming the dark biomass; the optima lie at 0.30 g/L in 15 cm and 0.15 g/L in 30 cm — the same areal loading of about 45 g/m². Computed for this module with k = 0.15 m²/g, surface irradiance 1,800 µmol·m⁻²·s⁻¹, saturation parameter 200 µmol·m⁻²·s⁻¹ and a maintenance coefficient of 0.28 µmol photon-equivalents per gram per second; those four assumptions set the result.

The right-hand panel contains the week's most useful result and it is not obvious. Once the culture is optically thick enough to absorb all the incident light — which happens at a few tenths of a gram per litre — adding more cells cannot raise areal productivity, because the photons were already all absorbed; it only redistributes them into a thinner surface layer, where high local irradiance wastes them to saturation, while the growing dark biomass consumes more in maintenance. The optimum therefore sits where those two losses balance, and it falls at a **fixed areal biomass loading of around 45 g/m² rather than a fixed concentration**: 0.30 g/L in a 15 cm pond, 0.15 g/L in a 30 cm one. That is why ponds are shallow and dilute, and it is the sentence to have ready when someone proposes to improve productivity by growing denser.

Now convert photons to biomass. A high-sun site delivers roughly 45 mol of photosynthetically active photons per square metre per day. At the theoretical minimum of about 10 photons per CO_2 fixed and about 24.6 g of biomass per C-mol, the photon-limited ceiling is around **110 g·m⁻²·day⁻¹**, and with the saturation and maintenance losses above, the model's productive photon use corresponds to roughly 50 g·m⁻²·day⁻¹. Reported outdoor productivities are 10 to 25 g·m⁻²·day⁻¹ — ten to twenty per cent of the photon ceiling. The gap is made of photoinhibition, which the saturating model does not contain at all: strong light damages photosystem II and the cell pays a continuous repair cost, so the response at midday is worse than saturation alone predicts. Then suboptimal temperature at the start and end of the day, night respiration, incomplete mixing between the lit and dark zones, and the fraction of the year the system runs at all.

In agricultural units, 20 g·m⁻²·day⁻¹ is 73 tonnes of dry biomass per hectare per year, against roughly 25 to 40 for a good land crop's dry matter. So phototrophs are genuinely two to three times more areally productive than crops — a real advantage, and nothing like the hundredfold sometimes implied — and the assumption in the model I would most want to measure is the specific attenuation coefficient k, because it sets the lit depth, the optimum loading and therefore everything downstream.

The two kinds of host, and what their genetics cost you

Cyanobacteria are prokaryotes: no organelles, photosynthesis on internal thylakoid membranes, a single circular chromosome, and in the model species natural competence and efficient homologous recombination, which makes targeted knockouts straightforward in principle. Eukaryotic microalgae have a nucleus, a chloroplast with its own genome and mitochondria, so a transgene can go into the nuclear genome or the chloroplast genome, and the two behave completely differently: chloroplast transformation gives high expression and homologous integration but no secretion and no eukaryotic processing, while nuclear transgenes integrate at random positions and are subject to silencing.

Two specific obstacles the week asks about. **Polyploidy**: *Synechocystis* carries multiple copies of its chromosome, so a knockout is not a knockout until every copy has segregated — which takes repeated restreaking under selection and must be verified by PCR showing complete loss of the wild-type allele. A partially segregated mutant behaves like a knockdown and has produced a quantity of irreproducible literature. **Silencing**: a nuclear transgene in *Chlamydomonas* is frequently silenced over generations, and the defences — codon optimisation, intron inclusion, screening many transformants, and using strains selected for expression — are all ways of managing a problem that has no clean solution.

Then the cycle time, which is what it actually costs a project. These organisms double in hours to a day rather than twenty minutes, transformation and segregation take weeks rather than days, and a growth experiment under light takes a week or two. A design–build–test cycle that is three days in *E. coli* is six to ten weeks in a phototroph, which by part 2 chapter 11's arithmetic is a factor of fifteen fewer hypotheses tested per year of funding. That is the single largest hidden cost of choosing a phototroph, and it should be stated in a project plan rather than discovered.

Three tool gaps to close before engineering a pathway in a host new to you: a characterised set of promoters and terminators with measured strengths in your conditions, since borrowed rankings do not transfer; a reliable selection and counter-selection system with markers that work and can be removed; and a validated transformation and segregation protocol with transformants per microgram quantified — which is the same first-three-experiments list as the yeasts, for the same reason.

Harvesting, which is where bulk products die

Cells are small, dilute, and close to neutrally buoyant, so getting them out of the water is a major cost term. Do the arithmetic once and you will never again read an algal-fuel projection the same way.

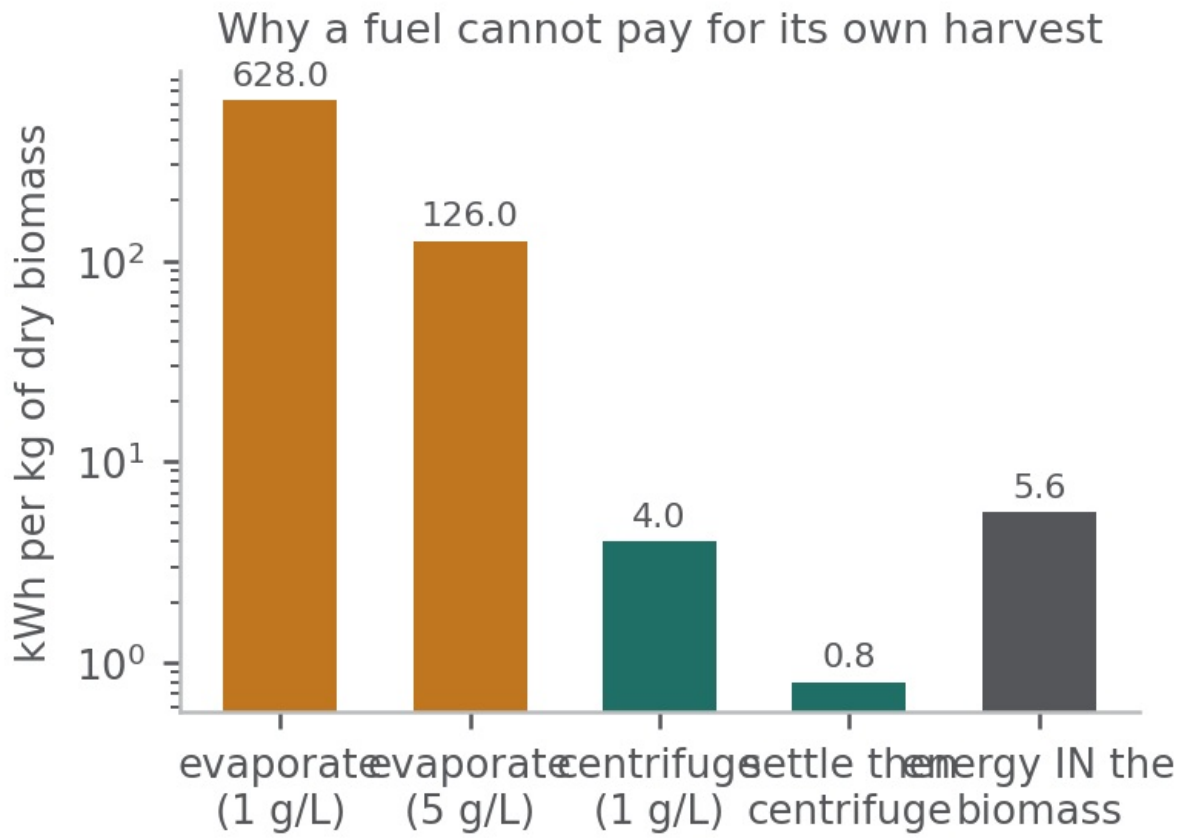


Figure 60.2 Energy per kilogram of dry biomass for different harvest routes, on a log scale, against the energy contained in the biomass itself. Evaporation is absurd by two orders of magnitude; centrifugation alone is of the same order as the biomass energy content; a staged route — flocculation or settling to concentrate, then centrifugation of a much smaller volume — is the only one that works. Computed from 2.26 MJ/kg for water evaporation and 1–8 kWh/m³ for centrifugation, with biomass taken at 20 MJ/kg.

At 1 g/L you must remove **1,000 litres of water per kilogram of dry biomass**. Evaporating it would take 2,260 MJ, or 628 kWh, against the roughly 20 MJ — 5.6 kWh — of chemical energy in the biomass: a factor of 113 against you. Centrifuging the whole volume at 1 to 8 kWh per cubic metre costs 1 to 8 kWh per kilogram, which is still of the same order as the energy in the product. Only a staged route works: flocculate or settle to concentrate the culture perhaps thirtyfold with very little energy, then centrifuge the small remaining volume, which brings the total under a kilowatt-hour per kilogram.

Connect that to part 4 chapter 49 and the conclusion is immediate. For a fuel — a product whose entire value is its energy content — a harvest step costing a comparable amount of energy cannot work, and no improvement in the strain changes it. For a product worth thousands per kilogram, the same harvest cost is irrelevant. **The dewatering arithmetic, not the biology, is what sorts phototroph products into viable and not.**

What a phototroph is actually for

Justified	Why	Not justified	The number that kills it
Astaxanthin, β-carotene and other pigments	Thousands of dollars per kilogram; the natural isomer commands a premium over the synthetic one; harvest cost is negligible against value	Transport fuels	Harvest energy of the same order as the fuel's energy content, before any conversion
Food and feed protein, and omega-3 oils	Value per kilogram in the tens of dollars; areal productivity two to three times a crop's; no agricultural land needed	Bulk commodity chemicals	A few dollars per kilogram against a dewatering and capital burden that a sugar fermentation does not carry
Carbon capture where capture itself is the point	The product can be low-value if the service is paid for	Bulk biomass for energy	The same as fuels, plus the land and water the system still needs

The week ends by asking you to write five honest sentences about your own microalgal work — glutathione and lipids — and the arithmetic above is how to do it without either dismissing it or overselling it. Something of this shape: the work was aimed at a product class where phototrophs can compete on value rather than on cost; the binding constraints were areal productivity and harvest, neither of which a strain improvement addresses; what a strain improvement can address

is content per unit biomass, which is exactly where glutathione and lipid engineering act; the honest reason to return to it would be a high-value product plus a co-product or service that pays for the harvest; and the honest reason not to is the design–build–test cycle time, which makes it a poor place to do iterative strain engineering. That is an answer with numbers in it, and it is also a much better interview answer than enthusiasm.

Host sheet — the phototrophs

Choose one when	The product is high-value and preferably accumulated in the biomass; when the CO_2 fixation itself is the point and is paid for; when a natural-source claim carries a premium.
Limiting substrate	Light, which cannot be stirred or fed. 15 mm of lit depth at 1 g/L; 95% of a 30 cm pond is dark.
Right unit	$g \cdot m^{-2} \cdot day^{-1}$. The optimum is an areal loading of about 45 g/m^2, so optimum density scales inversely with depth.
Realistic productivity	10–25 $g \cdot m^{-2} \cdot day^{-1}$ outdoors, 10–20% of the photon-limited ceiling, which is 36–90 $t \cdot ha^{-1} \cdot yr^{-1}$ against a crop's 25–40.
Harvest	1,000 L of water per kg of biomass at 1 g/L. Staged flocculation then centrifugation; evaporation is 113× the biomass energy content.
Tool cost	Polyploidy requiring full segregation in cyanobacteria; silencing and position effects for nuclear transgenes in algae; a DBTL cycle of six to ten weeks against three days in *E. coli*.
Wrong choice when	The product is a fuel or a bulk chemical, or the project needs many iterations of strain design.

Three things people get wrong

- ‘Grow the culture denser to raise productivity.’ Above full light absorption, more cells only waste photons to saturation and add maintenance; the optimum is an areal loading.
- ‘Phototrophs are vastly more productive than crops.’ Two to three times, areally — a real advantage and not an order of magnitude.
- ‘Harvesting is an engineering detail.’ At 1 g/L it is 1,000 litres of water per kilogram, and for a fuel it is the whole argument.

Self-check — answer on paper first

1. Compute the depth at which light falls to 1% of its surface value at 2 g/L, and say what it implies for mixing.

With $k = 0.15$ m^2/g, $kX = 300$ per metre, so the 1% depth is $\ln(100)/300 = 15$ mm. In a 30 cm pond that is 5% of the depth, so a cell spends about one twentieth of its time in usable light and the rest effectively dark. The implication is that mixing determines the *frequency* with which cells visit the lit layer, which matters because photosynthesis has a characteristic timescale: rapid light–dark cycling can use flashing light more efficiently than continuous dim light, which is the basis of the claimed flashing-light effect. Mixing cannot, however, increase the total photons available, so it redistributes rather than adds.

2. Why is the optimum biomass density halved when a pond is made twice as deep?

Because what the optimum fixes is the areal biomass loading — the grams of cells per square metre of illuminated surface — not the concentration. Once the culture absorbs all the incident light, areal productivity depends on how the fixed photon supply is distributed among a fixed areal quantity of cells, and the losses that create the optimum, photon waste at high local irradiance and maintenance of the dark biomass, both scale with that areal quantity. So $X \cdot L$ is the invariant, about 45 g/m^2 in the model here, and doubling L halves the optimal X.

3. A proposal claims 2 g/L/day in a 2 cm flat-panel reactor and projects it to a 100-hectare facility. What is wrong?

The volumetric figure was produced by the thinness of the panel and does not scale. Convert it: 2 g/L/day in a 2 cm path is 2 g per litre per day × 20 litres per square metre = 40 $g \cdot m^{-2} \cdot day^{-1}$, which is at the very top of what has been reported outdoors and is plausible only under optimal light and temperature — so the projection is assuming near-best-case areal performance across a hundred hectares and a full year. The questions to ask are what the areal figure is, over what period and with what light, and what the glazing and land capital cost per square metre is, since a flat-panel system at 100 hectares is an enormous capital number that a pond does not carry.

What to open a book for. Brock's phototrophy and microbial diversity chapters for the biology. Everything quantitative is in the literature the tracker points at: reviews of cyanobacteria as cell factories, of harvesting and dewatering energy, and of

areal productivity and light attenuation in photobioreactors, plus the genetic-tool literature for *Synechocystis* and *Chlamydomonas*. Part 3 chapter 48 of this module is the transport analogy and part 4 chapter 49 is where the harvest number belongs.

Chapter 61 · tracker week 61

The host table, cell-free systems, and reading the future tense

The deliverable of this week is one page: a table you could defend line by line in an interview. The skill it teaches is separating what has been demonstrated from what has been projected — which, in this field's reviews, is most of the text.

The host table

Here it is, filled as far as this module can defend. Cells marked with a dagger are where you should find a source rather than take my word: they are judgements rather than measurements, and the point of the exercise is to replace each with a citation of your own.

Host	Tool maturity	GRAS	Substrate range	Product localisation	Uniquely good at
E. coli	Highest	No	Sugars, glycerol, acetate; engineered C1	Cytoplasm or periplasm; poor secretion to medium	Being engineered fast — the shortest DBTL cycle in biology
S. cerevisiae	Highest among eukaryotes	Yes	Glucose, sucrose, maltose	Cytoplasm, organelles, or secreted	Eukaryotic pathways with P450s, plus the best eukaryotic toolkit
C. glutamicum	Good	Yes	Sugars; engineered on pentoses†	Secreted via native amino-acid exporters	Amino acids at million-tonne scale: native export plus tolerance
B. subtilis	Good	Yes	Sugars, starch	Secreted to medium across one membrane	Secreted enzymes at commercial titer; natural competence
Y. lipolytica	Moderate	Yes for several products†	Glucose, glycerol, acetate, alkanes, fatty acids	Stored as lipid bodies, or secreted	Lipids, oleochemicals and hydrophobic substrates
K. phaffii	Good for expression	Yes, food-enzyme history	Glycerol, methanol	Secreted, very clean background	High-density secretion of proteins — grams per litre
C. necator	Moderate	†	Sugars, fructose, plant oils, formate, H_2 + CO_2 + O_2	Stored as intracellular polymer	PHA, and autotrophic growth on gas
Cyanobacteria	Moderate; polyploidy complicates knockouts	Varies†	CO_2 and light	Cytoplasm or secreted†	Photoautotrophy with relatively tractable prokaryotic genetics
CHO cells	Industrialised but slow to engineer	n/a — clinical	Glucose and glutamine in defined medium	Secreted	Human-compatible glycosylation for therapeutic proteins

Two columns deserve comment because they decide more host arguments than the rest. **Product localisation** is the one most people leave out and the one that most often settles the question: secreted, retained or stored determines the downstream train, the titer ceiling by chapter 59's volume arithmetic, and whether toxicity is relieved. And **GRAS or regulatory history** is the only column that cannot be engineered on any timescale, which makes it decisive whenever the product goes into food, feed or a human.

The format for a host argument, which the phase 4 checkpoint asks for: name the property that decides it, name the alternative you rejected, and say what you gave up. "*K. phaffii*, because the product is a secreted haem protein and this host secretes at grams per litre with a clean background and a food-enzyme regulatory history; I rejected *E. coli*, which would have given a faster cycle and better tools but requires either periplasmic targeting or refolding and carries endotoxin that a food product cannot have; what I gave up is cycle time and the ability to use the promoter and assembly libraries I already know." An answer that says a host is standard or widely used is a default, not a choice, and interviewers are listening for exactly that distinction.

Constraining a phototroph model with photons

If a phototroph is on your list, the modelling step the week asks for is small and worth doing, because it makes chapter 60's arithmetic enter a flux model properly. A cyanobacterial genome-scale model has a photon exchange reaction — photons are a substrate like glucose — and its bound is normally set arbitrarily. Set it from your own attenuation calculation instead: compute the photons actually absorbed per gram of biomass per hour at your density and depth, which is the incident areal flux times the absorbed fraction divided by the areal biomass loading, and use that as the uptake bound. The model then predicts growth and product flux under a light limitation you can defend rather than one you assumed.

Two costs to note while you are there, because both are real and both are invisible in a stoichiometric model that has not

been told about them. RuBisCO is slow — a turnover number of a few per second, among the slowest central enzymes known — so a photoautotroph must invest an enormous fraction of its proteome in it, which by part 4 chapter 51's enzyme-constraint arithmetic is itself a growth limitation. And RuBisCO also fixes oxygen, so the carbon-concentrating mechanism exists to raise local CO_2 and suppress that, at a cost of roughly one ATP per CO_2 concentrated. Both belong in the model as constraints: a proteome cost and an ATP cost, respectively.

Cell-free systems: what removing viability buys

A cell-free system is a reaction containing the enzymes — as a lysate, as purified components, or as a transcription-translation system that makes them in situ — and not the cell. What that buys:

- **No competition for carbon.** There is no growth, so every substrate molecule is available to the pathway and yields can approach the stoichiometric ceiling rather than the ceiling minus biomass and maintenance, which parts 5's PHA chapter showed is a third of the gap.
- **No membrane.** Substrates, intermediates and cofactors can be added and measured directly, so the transport problems of part 2 chapter 14 and the compartment tax of chapter 58 disappear, and the concentrations are what you set rather than what the cell permits.
- **Tolerance of the intolerable.** A toxic intermediate or product cannot kill a system that is not alive, which is the single strongest reason to choose cell-free.
- **A short cycle.** Hours from DNA to measured function rather than days, which is why cell-free is used as a prototyping layer for parts and pathways before they are moved into cells.

What it costs: cofactors are no longer regenerated by a living metabolism, so ATP and NAD(P)H regeneration must be engineered into the reaction and is usually the binding constraint; the catalyst does not self-replicate, so enzyme preparation is a real cost per batch rather than a one-off; the system is unstable over long reactions as enzymes denature and phosphatases degrade cofactors; and scale is limited by the cost of lysate and of the energy source.

One product class where cell-free is genuinely the right answer: on-demand protein synthesis — vaccines, therapeutics and diagnostic reagents made at the point of need, including freeze-dried reactions rehydrated in the field — where the value per gram is high, the quantity is small, and the alternative is a cold chain. The second-best case is a short cascade with a toxic intermediate, where the tolerance argument dominates.

Reading the future tense

The last skill of this phase is reading a review sceptically, and it reduces to grammar. Mark every claim as **demonstrated** — it reports a result with a number, a host and a condition — or **projected**, which in practice means it contains *may*, *could*, *promises to*, *has the potential to*, *is expected to*, or *opens the way for*. Do this with a pen on a 2025 or 2026 review of self-driving biofoundries or AI-driven design and the proportion will surprise you; it is normal for the majority of the forward-looking sentences in a review of a fast field to be projections, and the failure mode is not the review's, it is quoting one of them in an interview as though it were a result.

Three current trends worth being able to discuss, with the demonstrated and projected parts separated. **Self-driving biofoundries**: demonstrated is high-throughput automated build and test with machine-learning-guided selection of the next round, in specific well-instrumented cases; projected is autonomy across the whole loop including hypothesis generation. **Enzyme-constrained and omics-integrated models**: demonstrated is that a proteome constraint reproduces overflow metabolism and improves growth prediction, as part 4 chapter 51 showed; projected is reliable prediction of engineered-strain phenotypes, which remains poor. **C1 and CO_2 valorisation**: demonstrated is gas fermentation at commercial scale for ethanol and growth on methanol and formate in engineered strains; projected is cost parity with sugar, which chapter 57 of part 5 showed to depend on an electricity price that mostly does not exist yet.

And the honest version of the question "which trend would you want to work inside": the one where the bottleneck matches what you are good at. For a chemical engineer who has done the arithmetic in this module, that is the modelling-and-constraints trend and the C1 and gas trend, because both are limited by transport, thermodynamics and cost accounting rather than by molecular biology, and both have more people who can clone than people who can close a balance.

The phase 4 checkpoint

1. Reconstruct artemisinin, 1,4-BDO and LanzaTech from memory in the seven-field format.

Artemisinin. Host *S. cerevisiae*; target artemisinic acid as a precursor to an antimalarial; pathway mevalonate plus amorphadiene synthase plus a plant P450 with its reductase; tools yeast genetics, promoter engineering, P450 expression; TRY 25 g/L with yield far below the 0.29 g/g ceiling and nobody minding; bottleneck P450 activity and acetyl-

CoA plus ATP supply; commercial status produced, then undercut by agricultural supply.
1,4-BDO. Host *E. coli*; target a non-natural four-carbon diol; pathway found computationally from succinyl-CoA and α-ketoglutarate; tools genome-scale modelling, growth-coupled strain design, multiplexed editing; TRY 18 g/L in the paper and above 100 g/L commercially; bottleneck precursor supply and redox under anaerobic conditions; commercial status produced at scale into polymers.
LanzaTech. Host *Clostridium autoethanogenum*; target ethanol and C2–C4 products; pathway native Wood–Ljungdahl; feedstock steel-mill off-gas containing CO, CO_2 and H_2; TRY driven by gas transfer rather than by pathway; bottleneck ATP — acetogen energetics leave almost no margin, which limits products to those close to acetyl-CoA; commercial status operating since 2018.

2. Defend your capstone host choice against one named alternative.

No fixed answer, but the structure is fixed: the decisive property, the alternative rejected, and what was given up. Name a measurable property — P450 support, native lipid flux, secretion titer, gas uptake rate, GRAS status, transformants per microgram — not a reputation.

3. Mark every forward-looking claim in a 2025 review as demonstrated or projected.

Check the marking against the tense. Demonstrated claims report a result with a number; projected claims use may, could, promises to, or has the potential to. Reviews in this field carry a great deal of the second kind, and quoting one as though it were the first is the fastest way to lose credibility in an interview.

What to open a book for. No book at all this week. A 2025 or 2026 review on AI-driven biofoundries or systems metabolic engineering, read with a pen for the demonstrated-versus-projected exercise, and a cell-free metabolic engineering review. The host table is your own document and should end the week with a citation in every cell that currently carries a dagger.

Closing notes to part 6

The four weeks in one page

Week 58. Yeast gives you folding, secretion, low pH, GRAS status and organelles. Its central fact is that acetyl-CoA is mitochondrial and your pathway is cytosolic, so cytosolic supply is the first thing to engineer — at one ATP per unit through ATP-citrate lyase or two through the pyruvate dehydrogenase bypass. The lipid ceiling is 0.356 g/g on carbon alone and 0.275 once NADPH is paid for, with 15 ATP per palmitate on top. Compartmentalisation buys a hundredfold concentration gain and charges a transporter for everything that crosses — about 2.8×10^{-8} mol·m^{-2}·s^{-1} against a carrier's 1.7×10^{-7}, so it works with roughly sixfold margin and only with a dedicated carrier.

Week 59. One membrane instead of two is the whole argument: secretion crosses one barrier, there is no endotoxin, and the downstream train loses two unit operations. The intracellular ceiling is not a fraction of a gram per litre but scales with cell density — 20 g/L at 50 gDCW/L, 40 at 100 — which is why intracellular products force high-cell-density operation. Lysine's ceiling moves from 0.81 to 0.86 mol/mol with a transhydrogenase, a cofactor device rather than a carbon one, and export work grows only logarithmically with titer: 4.8 to 7.6 kJ/mol across 50 to 150 g/L.

Week 60. Light reaches 15 mm at 1 g/L, so 95% of a pond is dark, and the unit is g·m^{-2}·day^{-1}. Above full absorption more cells cannot help, so the optimum is an areal loading of about 45 g/m² — 0.30 g/L in 15 cm, 0.15 in 30 cm. Realistic output is 10–25 g·m^{-2}·day^{-1}, ten to twenty per cent of the photon ceiling and two to three times a crop areally. And 1,000 litres of water per kilogram of biomass is what sorts the viable products from the rest.

Week 61. One table, with product localisation and regulatory history as the two columns that decide most arguments. Cell-free buys freedom from growth competition, membranes and toxicity at the cost of cofactor regeneration and a catalyst that does not replicate. And in a review, mark every claim demonstrated or projected before you quote it.

On the numbers in this part

The lipid ceiling, the peroxisome flux density and carrier capacity, the intracellular-titer table, the export work, the light attenuation depths, the areal optimum and the dewatering energies were all computed from the assumptions printed beside them. Four of those assumptions carry most of the weight and should be the first things you replace with measurements: the NADPH source in the lipid calculation, which is the entire difference between my 0.275 g/g and the literature's 0.30 to 0.33; the cell volume per gram of dry weight, taken here as 2 mL; the specific light attenuation coefficient k, taken as 0.15 m²/g, which sets the lit depth and the areal optimum together; and the maintenance coefficient in the pond model, which is what creates the optimum at all. Host properties in the tables that are judgements rather than measurements are marked with a dagger, and the week's exercise is to replace each with a source.

Where this leaves the module

Six parts now exist. Part 1 covered the biology foundations of weeks 1 to 10 as a spine beside the textbooks. Part 2 did the modelling and control framework of weeks 11 to 19 at twice that density. Part 3 covered in situ removal and immobilisation, parts 4 the economics, validation, omics and host defences, part 5 the five product classes with a constraint sheet each, and this part the hosts against those sheets. Together they cover weeks 1 to 19 and 45 to 61 — 36 of the tracker's 66 weeks, and most of the weeks with nothing behind them.

What remains unwritten, in tracker order: weeks 20 to 44, the technique and process phases, where the sources are protocols, software documentation and reviews that are genuinely good and genuinely specific — assembly and editing, screening, analytics, modelling software, fermentation and scale-up; and weeks 62 to 66, the publishing, proposal and application weeks, which are about how the profession works rather than about metabolic engineering. The technique weeks are the ones where a written module adds least, because a protocol you can run beats a chapter about it; the professional weeks are the ones where it might add a surprising amount, since nobody writes that down for engineers coming in from another discipline. If you want more, that is where I would go next, and I would ask which of the two you would actually use.